CHINESE REMAINDER THEOREM

$$x \cong a_1 (\text{mod } m_1)$$
$$x \cong a_2 (\text{mod } m_2)$$
$$x \cong a_3 (\text{mod } m_3)$$

$$\gcd(m_1, m_2) = \gcd(m_2, m_3) = \gcd(m_3, m_1) = 1$$

$$x = (M_1 X_1 a_1 + M_2 X_2 a_2 + M_3 X_3 a_3 \ldots\ldots M_n X_n a_n) \text{ mod } M.$$

$M = m_1 \times m_2 \times m_3$

$M_1 = m_2 m_3$ $\qquad M_1 X_1 \cong 1 \text{ mod } m_1$

$M_2 = m_1 m_3$ $\qquad M_2 X_2 \cong 1 \text{ mod } m_2$

$M_3 = m_1 m_2$ $\qquad M_3 X_3 \cong 1 \text{ mod } m_3$

Q. $x \cong 1 (\text{mod } 5)$
$x \cong 1 (\text{mod } 7)$
$x \cong 3 (\text{mod } 11)$

$M = m_1 \times m_2 \times m_3$

$M = 5 \times 7 \times 11 = 385$

$M_1 = m_2 \times m_3$

$M_1 = 7 \times 11 = 77$

$M_2 = m_1 \times m_3$

$M_2 = 5 \times 11 = 55$

$M_3 = m_1 \times m_2$

$M_3 = 5 \times 7 = 35$

$\gcd(m_1, m_2)$
$\gcd(5, 7)$
$7 = 5 \times 1 + 2$
$5 = 2 \times 2 + 1$
$2 = 1 \times 2 + 0$

$\gcd(m_2, m_3)$
$\gcd(7, 11)$
$11 = 7 \times 1 + 4$
$7 = 4 \times 1 + 3$
$4 = 3 \times 1 + 1$
$3 = 1 \times 3 + 0$

$\gcd(m_1, m_3)$
$\gcd(5, 11)$
$11 = 5 \times 2 + 1$
$5 = 1 \times 5 + 0$

Calculate X_1:

$M_1 X_1 \cong 1 \bmod m_1$

$77 X_1 \cong 1 \bmod 5$

$77 X_1 \pmod 5 \cong 1$

$2 X_1 \pmod 5 \cong 1$

$\boxed{X_1 = 3}$

Calculate X_2:

$M_2 X_2 \cong 1 \bmod m_2$

$55 X_2 \cong 1 \bmod 7$

$55 X_2 \pmod 7 \cong 1$

$6 X_2 \pmod 7 \cong 1$

$\boxed{X_2 = 6}$

Calculate X_3:

$M_3 X_3 \cong 1 \bmod m_3$

$35 X_3 \cong 1 \bmod 11$

$35 X_3 \pmod{11} \cong 1$

$2 X_3 \pmod{11} \cong 1$

$\boxed{X_3 = 6}$

Verification:

$36 \bmod 5 = 1$ ✓

$36 \bmod 7 = 1$ ✓

$36 \bmod 11 = 3$ ✓

$X = (M_1 X_1 a_1 + M_2 X_2 a_2 + M_3 X_3 a_3) \bmod M$

$X = (77 \times 3 \times 1 + 55 \times 6 \times 1 + 35 \times 6 \times 3) \bmod 385$

$X = (231 + 330 + 630) \bmod 385$

$X = 1191 \bmod 385$

$\boxed{X = 36}$

LINEAR CONGRUENTIAL GENERATORS

m	the modulus	$m > 0$
a	the multiplier	$0 < a < m$
c	the increment	$0 \leq c < m$
x_0	the starting value or seed	$0 \leq x_0 < m$

Iterative Equation:: $X_{n+1} = (aX_n + C) \bmod m$.

eg: $a=3$, $c=5$, $m=17$, $X_0=2$.

1.) $X_1 = (aX_0+c) \bmod m \Rightarrow (3\times2+5) \bmod 17 = 11$
2.) $X_2 = (aX_1+c) \bmod m \Rightarrow (3\times11+5) \bmod 17 = 4$
3.) $X_3 = (aX_2+c) \bmod m \Rightarrow (3\times4+5) \bmod 17 = 0$
4.) $X_4 = (aX_3+c) \bmod m \Rightarrow (3\times0+5) \bmod 17 = 5$
5.) $X_5 = (aX_4+c) \bmod m \Rightarrow (3\times5+5) \bmod 17 = 3$
6.) $X_6 = (aX_5+c) \bmod m \Rightarrow (3\times3+5) \bmod 17 = 14$
7.) $X_7 = (aX_6+c) \bmod m \Rightarrow (3\times14+5) \bmod 17 = 13$
8.) $X_8 = (aX_7+c) \bmod m \Rightarrow (3\times13+5) \bmod 17 = 10$
9.) $X_9 = (aX_8+c) \bmod m \Rightarrow (3\times10+5) \bmod 17 = 1$
10.) $X_{10} = (aX_9+c) \bmod m \Rightarrow (3\times1+5) \bmod 17 = 8$
11.) $X_{11} = (aX_{10}+c) \bmod m \Rightarrow (3\times8+5) \bmod 17 = 12$
12.) $X_{12} = (aX_{11}+c) \bmod m \Rightarrow (3\times12+5) \bmod 17 = 7$
13.) $X_{13} = (aX_{12}+c) \bmod m \Rightarrow (3\times7+5) \bmod 17 = 9$
14.) $X_{14} = (aX_{13}+c) \bmod m \Rightarrow (3\times9+5) \bmod 17 = 15$
15.) $X_{15} = (aX_{14}+c) \bmod m \Rightarrow (3\times15+5) \bmod 17 = 16$
16.) $X_{16} = (aX_{15}+c) \bmod m \Rightarrow (3\times16+5) \bmod 17 = 2$

Resulting Sequence: 11, 4, 0, 5, 3, 14, 13, 10, 1, 8, 12, 7, 9, 15, 16, 2

BLUM BLUM SHUB

1. $p, q \rightarrow 2$ Prime no.s
   $p = q$ divided by 4 so remainder $= 3$
   $p = q = 3 \pmod 4$
2. $n = p \times q$
3. $s =$ relatively prime to n.
4. $x_0 = s^2 \bmod n$.
5. for $i = 1$ to k ($k =$ no. of random no.s).
6. $x_i = (x_{i-1})^2 \bmod n$.
7. $B_i = x_i \bmod 2$.

Q: $p = 7$, $q = 11$, Find 2 sequence bit

$n = p \times q \Rightarrow n = 7 \times 11 = 77$ $\quad$ $i = 1$ to 2 $\therefore k = 2$

$s = \gcd(s, 77) = 1 \quad \therefore s = 12$

$x_0 = s^2 \bmod n \Rightarrow x_0 = (12)^2 \bmod 77 \Rightarrow \boxed{x_0 = 67}$

$x_1 = (x_0)^2 \bmod n \Rightarrow x_1 = (67)^2 \bmod 77$

$\boxed{x_1 = 23}$

$B_1 = x_1 \bmod 2 \Rightarrow B_1 = 23 \bmod 2$

$\boxed{B_1 = 1}$

$x_2 = (x_1)^2 \bmod n \Rightarrow x_2 = (23)^2 \bmod 77$

$\boxed{x_2 = 67}$

$B_2 = x_2 \bmod 2 \Rightarrow B_2 = 67 \bmod 2$

$\boxed{B_2 = 1}$

Sequence Bit: 11

EUCLIDEAN ALGORITHM

gcd(60, 24) $\quad$ ∵ n = positive number, r = remainder.

$$60 = 24 \times \overset{n+r}{[redacted]}$$

$60 = 24 \times 2 + 12$ $\quad$ gcd(60, 24) = 12

$24 = 12 \times 2 + 0$

gcd(12, 33)

$33 = 12 \times 2 + 9$

$12 = 9 \times 1 + 3$ $\quad$ gcd(12, 33) = 3

$9 = 3 \times 3 + 0$

FERMAT'S LITTLE THEOREM

If p is a prime number and 'a' is a positive integer not divisible by p then,

$$a^{p-1} = 1 \pmod{p}.$$

Q: Does fermat's theorem hold true for p=5 and a=2?

$a^{p-1} = 1 \pmod{p}$.

$2^{5-1} = 1 \pmod{5}$

$2^{4} = 1 \pmod{5}$

$16 = 1 \pmod{5}$

$16 \bmod 5 = 1$

Therefore, fermat's theorem holds true for p=5 and a=2.

EULER'S THEOREM

For every positive integer 'a' & 'n', which are said to be relatively prime then,

$$a^{\phi(n)} \equiv 1 \bmod n.$$

$\phi(n) \rightarrow$ Totient Function.

$n = 6$

$\phi(6) = 1, 2, 3, 4, 5$ $\quad$ $\because$ gcd (number, 6)

$\boxed{\phi(6) = 2}$ $\{1, 5\}$

Example 1:- n = Prime

$\phi n = n - 1$

$n = 7$

$\phi(n) = 7 - 1$

$\boxed{\phi(n) = 6}$

Example 2:- n = odd

$\phi n = (p-1)(q-1)$

$n = 15 \Rightarrow \underset{p}{3} \times \underset{q}{5}$

$\phi(n) = (3-1)(5-1)$

$\boxed{\phi(n) = 8}$

Example 3:- n = Even

$\phi(n) = n\left(1 - \frac{1}{p}\right)$

$n = 8 \Rightarrow \underset{p}{2}^3$

$\phi(n) = 8\left(1 - \frac{1}{2}\right)$

$\boxed{\phi(n) = 4}$

DIFFIE-HELLMAN KEY EXCHANGE

- α should be the premitive root of q and $\alpha < q$
- q should be a prime number.

$$n = \alpha^n \bmod q$$

Q1. $q = 11$

$\alpha^n \bmod q$

- $2^1 \bmod 11 = 2$
- $2^2 \bmod 11 = 4$
- $2^3 \bmod 11 = 8$
- $2^4 \bmod 11 = 5$
- $2^5 \bmod 11 = 10$
- $2^6 \bmod 11 = 9$
- $2^7 \bmod 11 = 7$
- $2^8 \bmod 11 = 3$
- $2^9 \bmod 11 = 6$
- $2^{10} \bmod 11 = 1$

- $\underbrace{X_A = 4, \quad X_B = 6}_{\text{given}} \rightarrow$ secret numbers

Public Key Generation:-

(For Alice) $Y_A = \alpha^{X_A} \bmod q$

$Y_A = 2^4 \bmod 11$

$\boxed{Y_A = 5}$

(For Bob) $Y_B = \alpha^{X_B} \bmod q$

$Y_B = 2^6 \bmod 11$

$\boxed{Y_B = 9}$

Private Key Generation:-

$K = (Y_B)^{X_A} \bmod q$ (For Alice)

$K = (9)^4 \bmod 11$

$\boxed{K = 5}$

(For Bob)

$K = (Y_A)^{X_B} \bmod q$

$K = (5)^6 \bmod 11$

$\boxed{K = 5}$

Always same

Q2. $q = 7$

$\alpha^n \bmod q$

$2^1 \bmod 7 = 2$

$2^2 \bmod 7 = 4$

$2^3 \bmod 7 = 1$

$2^4 \bmod 7 = 2$ x Distinct Values

$3^1 \bmod 7 = 3$

$3^2 \bmod 7 = 2$

$3^3 \bmod 7 = 6$

$3^4 \bmod 7 = 4$

$3^5 \bmod 7 = 5$

$3^6 \bmod 7 = 1 \rightarrow$ All distinct values.

$X_A = 6$, $X_B = 4$

Public Key Generation:-

(For Alice) $Y_A = \alpha^{X_A} \bmod q$

$Y_A = 3^6 \bmod 7$

$\boxed{Y_A = 1}$

(for Bob) $Y_B = \alpha^{X_B} \bmod q$

$Y_B = 3^4 \bmod 7$

$\boxed{Y_B = 4}$

Private Key Generation:-

(For Alice) $K = (Y_B)^{X_A} \bmod q$

$K = 4^6 \bmod 7$

$\boxed{K = 1}$

(For Bob) $K = (Y_A)^{X_B} \bmod q$

$K = 1^4 \bmod 7$

$\boxed{K = 1}$

Same.

RSA

Ex: $p=3$, $q=5$

- Key Generation:

Calculate n;

$$n = p \times q$$
$$n = 3 \times 5 = 15$$

Calculate $\phi(n)$;

$$\phi(n) = (p-1)(q-1)$$
$$\phi(n) = (3-1)(5-1) = 8$$

Select integer e;

$$\gcd(\phi(n), e) = 1$$
$$\gcd(8, 3) = 1$$
$$8 = 3 \times 2 + 2$$
$$3 = 2 \times 1 + 1$$
$$2 = 1 \times 2 + 0$$

so, $e = 3$

Calculate d;

$$de = 1 \bmod \phi(n)$$
$$d(3) = 1 \bmod 8$$
$$d \times 3 \bmod 8 = 1$$
$$3 \times 3 \bmod 8 = 1$$
$$9 \bmod 8 = 1$$
$$\boxed{1 = 1}$$

so, $d = 3$

Public Key $= \{e, n\} = \{3, 15\}$
Private Key $= \{d, n\} = \{3, 15\}$

- Encryption:

Plaintext = P = 8 (Given)

$C = P^e \mod n$

$C = 8^3 \mod 15$

$\boxed{C = 2}$

- Decryption:

Ciphertext = C = 2 (Given)

$P = C^d \mod n$

$P = 2^3 \mod 15$

$\boxed{P = 8}$

RC-4

Q, 3 bit of Plaintext

$S = 0$ to 7

$K = [1\ 2\ 3\ 6]$

$P = [1\ 2\ 2\ 2] \rightarrow 3$ bit

$$T = \begin{array}{c} \begin{matrix} 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 \end{matrix} \\ \begin{bmatrix} 1 & 2 & 3 & 6 & 1 & 2 & 3 & 6 \end{bmatrix} \end{array} \rightarrow \text{Repeat } K \text{ once.}$$

$$S = \begin{array}{c} \begin{bmatrix} 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 \end{bmatrix} \\ \begin{matrix} 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 \end{matrix} \end{array}$$

$j = (j + S[i] + T[i]) \bmod 8$

• (KSA) swap $(S[i], S[j])$

First Iteration: (For $i=0, j=0$).

$j = (0+0+1) \bmod 8$

$j = 1 \bmod 8$

$\boxed{j=1}$

Swap $(S[0], S[1])$

$$S = \begin{array}{c} \begin{bmatrix} 1 & 0 & 2 & 3 & 4 & 5 & 6 & 7 \end{bmatrix} \\ \begin{matrix} 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 \end{matrix} \end{array}$$

• Second Iteration: (For $i=1, j=1$)

$j = (1+0+2) \bmod 8$

$\boxed{j=3}$

Swap $(S[1], S[3])$

$$S = \begin{array}{c} \begin{bmatrix} 1 & 3 & 2 & 0 & 4 & 5 & 6 & 7 \end{bmatrix} \\ \begin{matrix} 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 \end{matrix} \end{array}$$

Third Iteration: (For $i=2, j=3$)

$j = (3 + 2 + 3) \bmod 8$

$\boxed{j=0}$

Swap $(S[2], S[0])$

$$S = \begin{array}{c} \begin{bmatrix} 2 & 3 & 1 & 0 & 4 & 5 & 6 & 7 \end{bmatrix} \\ \begin{matrix} 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 \end{matrix} \end{array}$$

Fourth Iteration:- (For $i=3$, $j=0$)

$j=(0+0+6) \bmod 8$

$\boxed{j=6}$

Swap$(S[3], S[6])$

$S=[2\ 3\ 1\ 6\ 4\ 5\ 0\ 7]$

0 1 2 3 4 5 6 7

Fifth Iteration:- (For $i=4$, $j=6$)

$j=(6+4+1) \bmod 8$

$\boxed{j=3}$

Swap$(S[4], S[3])$

$S=[2\ 3\ 1\ 4\ 6\ 5\ 0\ 7]$

0 1 2 3 4 5 6 7

Sixth Iteration:- (For $i=5$, $j=3$)

$j=(3+5+2) \bmod 8$

$\boxed{j=2}$

Swap$(S[5], S[2])$

$S=[2\ 3\ 5\ 4\ 6\ 1\ 0\ 7]$

0 1 2 3 4 5 6 7

Seventh Iteration:- (For $i=6$, $j=2$)

$j=(2+0+3) \bmod 8$

$\boxed{j=5}$

Swap$(S[6], S[5])$

$S=[2\ 3\ 5\ 4\ 6\ 0\ 1\ 7]$

0 1 2 3 4 5 6 7

Eigth Iteration:- (For $i=7, j=5$)

$j=(5+7+6) \bmod 8$

$\boxed{j=2}$

Swap$(S[7], S[2])$

$S=[2\ 3\ 7\ 4\ 6\ 0\ 1\ 5]$

Final S after KSA

$$S=\begin{array}{cccccccc} [2 & 3 & 7 & 4 & 6 & 0 & 1 & 5] \\ 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 \end{array}$$

• (Simplified Stream Generator) → 4 iterations for 3-bit plaintext

→ $i=(i+1) \bmod 8$

→ $j=(j+S[i]) \bmod 8$

Swap$(S[i], S[j])$

→ $t=(S[i]+S[j] \bmod 8$

$K=S[t]$

First Iteration:- (For $i=0, j=0$)

$i=(0+1) \bmod 8$

$\boxed{i=1}$

$j=(0+3) \bmod 8$

$\boxed{j=3}$

Swap$(S[1], S[3])$

$$S=\begin{array}{cccccccc} [2 & 4 & 7 & 3 & 6 & 0 & 1 & 5] \\ 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 \end{array}$$

$t=(4+3) \bmod 8$

$\boxed{t=7}$

$K=S(7)$

$\boxed{K=5}$

Second Iteration:- (For $i=1, j=3$)

$i=(1+1) \bmod 8$

$\boxed{i=2}$

$j=(3+7) \bmod 8$

$\boxed{j=2}$

Swap$(S[2], S[2])$

$$S=\begin{array}{cccccccc} [2 & 4 & 7 & 3 & 6 & 0 & 1 & 5] \\ 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 \end{array}$$

$t=(7+7) \bmod 8$

$\boxed{t=6}$

$K=S(6)$

$\boxed{K=1}$

Third Iteration:- (For $i=2, j=2$)

$i=(2+1) \bmod 8$

$\boxed{i=3}$

$j=(2+3) \bmod 8$

$\boxed{j=5}$

Swap$(S[3], S[5])$

$$S=\begin{array}{cccccccc} [2 & 4 & 7 & 0 & 6 & 3 & 1 & 5] \\ 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 \end{array}$$

$t=(0+3) \bmod 8$

$\boxed{t=3}$

$K=S(3)$

$\boxed{K=0}$

Fourth Iteration:- (For $i=3, j=5$)

$i=(3+1) \bmod 8$

$\boxed{i=4}$

$j=(5+6) \bmod 8$

$\boxed{j=3}$

Swap$(S[4], S[3])$

$$S=\begin{array}{cccccccc} [2 & 4 & 7 & 6 & 0 & 3 & 1 & 5] \\ 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 \end{array}$$

$t=(0+6) \bmod 8$

$\boxed{t=6}$

$K=S(6)$

$\boxed{K=1}$

After final iteration

$KS=[5\ 1\ 0\ 1]$

• (Cipher Text)

$CT = PT \oplus KS$

$PT=[1\ 2\ 2\ 2]$

$KS=[5\ 1\ 0\ 1]$

PT	001	010	010	010
⊕KS	101	001	000	001
	100	011	010	011
	4	3	2	3

$CT=[4\ 3\ 2\ 3]$

AES ALGORITHM

Initial Key (128 bits):

01110011 01100001 01110100 00110 ....

W_0	W_1	W_2	W_3
73	73	69	72
61	68	73	69
74	63	62	6E
69	6A	6F	67

$W_3 \rightarrow g$

W_4	W_5	W_6	W_7
8B	F8	91	E3
FE	97	E4	8D
F1	92	F0	9E
29	43	2C	4B

$W_4 = W_0 \oplus g(W_3)$

$W_5 = W_4 \oplus W_1$

$W_6 = W_5 \oplus W_2$

$W_7 = W_6 \oplus W_3$

Find g: W3 ko 1-1 kia hai; Rotword pe S-box lagaya hai

W_3	Rotword	Subword	R(con)
72	69	F9	F8
69	6E	9F	9F
6E	67	85	85
67	72	40	40

$\left.\begin{matrix} F8 \\ 9F \\ 85 \\ 40 \end{matrix}\right\} g(W_3)$

- R(con)

Subword $\oplus$ R_1

$F9 \oplus 01$ $9F \oplus 00$ $85 \oplus 00$ $40 \oplus 00$

F8   9F   85   40

(isko hum pehle binary main convert karinge phir $\oplus$ karinge)

R_1	R_2	R_3	R_4	R_5	R_6	R_7	R_8	R_9	R_{10}
01	02	04	08	10	20	40	80	1B	36
00	00	00	00	00	00	00	00	00	00
00	00	00	00	00	00	00	00	00	00
00	00	00	00	00	00	00	00	00	00

$W_4 = W_0 \oplus g(W_3)$

$(W_0) =$ 0111 0011 0110 0001 0111 0100 0110 1001

$\oplus\, g(W_3) =$ 1111 1000 1001 1111 1000 0101 0100 0000

1000 1011 1111 1110 1111 0001 0010 1001

$W_4 =$ 8 B F E F 1 2 9

$W_4 =$ 8B FE F1 29

$W_5 = W_4 \oplus W_1$

$(W_4) =$ 1000 1011 1111 1110 1111 0001 0010 1001

$\oplus\, (W_1) =$ 0111 0011 0110 1001 0110 0011 0110 1010

1111 1000 1001 0111 1001 0010 0100 0011

F 8 9 7 9 2 4 3

$W_5 =$ F8 97 92 43

$W_6 = W_5 \oplus W_2$

$(W_5) =$ 1111 1000 1001 0111 1001 0010 0100 0011

$\oplus\, (W_2) =$ 0110 1001 0111 0011 0110 0010 0110 1111

1001 0001 1110 0100 1111 0000 0010 1100

9 1 E 4 F 0 2 C

$W_6 =$ 91 E4 F0 2C

$W_7 = W_6 \oplus W_3$

$(W_6) =$ 1001 0001 1110 0100 1111 0000 0010 1100

$\oplus\, (W_3) =$ 0111 0010 0110 1001 0110 1110 0110 0111

1110 0011 1000 1101 1001 1110 0100 1011

E 3 8 D 9 E 4 B

$W_7 =$ E3 8D 9E 4B

$r_1 = 1\times1$	$r_5 = 1\times2$	$r_9 = 1\times3$	$r_{13} = 1\times4$
$r_2 = 2\times1$	$r_6 = 2\times2$	$r_{10} = 2\times3$	$r_{14} = 2\times4$
$r_3 = 3\times1$	$r_7 = 3\times2$	$r_{11} = 3\times3$	$r_{15} = 3\times4$
$r_4 = 4\times1$	$r_8 = 4\times2$	$r_{12} = 4\times3$	$r_{16} = 4\times4$

Step 1: Byte Substitution

EA	04	65	85
83	45	5D	96
5C	33	98	B0
F0	2D	AD	C5

S-box apply →

87	F2	4D	97
EC	6E	4C	90
4A	C3	46	E7
8C	D8	95	A6

Step 2: Shift Rows

Row 1 → No shift
Row2 → 1 L shift
Row3 → 2 L-Shift
Row4 → 3 L-Shift

87	F2	4D	97
6E	4C	90	EC
46	E7	4A	C3
A6	8C	D8	95

Step 3: Mix Column $(x^7+x^6+x^5+x^4+x^3+x^2+x+1)$

$$\begin{bmatrix} 02 & 03 & 01 & 01 \\ 01 & 02 & 03 & 01 \\ 01 & 01 & 02 & 03 \\ 03 & 01 & 01 & 02 \end{bmatrix} * \begin{bmatrix} 87 & F2 & 4D & 97 \\ 6E & 4C & 90 & EC \\ 46 & E7 & 4A & C3 \\ A6 & 8C & D8 & 95 \end{bmatrix}$$

$r_1 = (02\times87) \oplus (03\times6E) \oplus (01\times46) \oplus (01\times A6)$

$02\times87 = [(00000010)(10000111)]$ $\quad \because 02 = x,\ 87 = x^7+x^2+x+1$

$(02\times87) = x(x^7+x^2+x+1) \Rightarrow (02\times87) = x^8+x^3+x^2+x$

Use irreducible Polynomial Theorem, $GF(2^3)$

$x^8 = x^4+x^3+x+1$

$(02\times87) = x^4+\cancel{x^3}+\cancel{x}+1+\cancel{x^3}+x^2+\cancel{x}$

$(02\times87) = x^4+x^2+1$

$(02\times87) = 0001\ 0101$

$(03 \times 6E) = [(0000\ 0011)(0110\ 1[illegible]10)]$ ∵ $03 = x+1$, $6E = x^6+x^5+x^3+x^2+x$

$(03 \times 67) = (x+1)(x^6+x^5+x^3+x^2+x) \Rightarrow x^7 + \cancel{x^6} + x^4 + \cancel{x^3} + \cancel{x^2} + \cancel{x^6} + x^5 + \cancel{x^3} + \cancel{x^2} + x$

$\Rightarrow x^7 + x^5 + x^4 + x$

$(03 \times 67) = 1011\ 0010$

$(01 + 46) = [(00000001)(0100[illegible]10)]$ ∵ $01 = 1$, $46 = x^6 + x^2 + x$

$(01 \times 46) = 1(x^6 + x^2 + x) \Rightarrow x^6 + x^2 + x$

$(01 \times 46) = 0100[illegible]0110$

$(01 \times A6) = [(00000001)(10100110)]$ ∵ $01 = 1$, $A6 = x^7 + x^5 + x^2 + x$

$(01 \times A6) = 1(x^7 + x^5 + x^2 + x) \Rightarrow x^7 + x^5 + x^2 + x$

$(01 \times A6) = 10100110$

Perform XOR on all.

(02×87)	0 0 0 1 0 1 0 1
(03×67)	1 0 1 1 0 0 1 0
(01×46)	0 1 0 0 0 1 1 0
$(01 \times A6)$ ⊕	1 0 1 0 0 1 1 0
	0 1 0 0 0 1 1 1
	4 7

$\boxed{r_1 = 47}$

Step 4:: Add Round Key

New state ⊕ Round Key = Another new state.

(Pehla Column of new state ⊕ Pehla column of round key, Same with others).